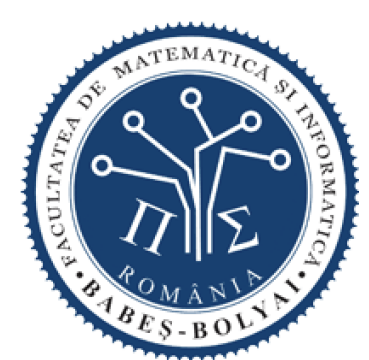




Operations Research

03 – Simplex method

Mihai Suci

Faculty of Mathematics and Computer Science

Simplex method when some constraints are not “ $\leq$ ” constraints

- We employ a mathematical “trick” to jumpstart the problem by adding **artificial variables** to the equations.

Simplex method when some constraints are not “ $\leq$ ” constraints (cont.)

Example:

$$\text{Max } 16x_1 + 15x_2 + 20x_3 - 18x_4$$

ST

$$2x_1 + x_2 + 3x_3 \leq 3000 \quad [1]$$

$$3x_1 + 4x_2 + 5x_3 - 60x_4 \leq 2400 \quad [2]$$

$$x_4 \leq 32 \quad [3]$$

$$x_2 \geq 200 \quad [4]$$

$$x_1 + x_2 + x_3 \geq 800 \quad [5]$$

$$x_1 - x_2 - x_3 = 0 \quad [6]$$

$$x_j \geq 0 \text{ for all } j$$

Simplex method when some constraints are not “ $\leq$ ” constraints (cont.)

Example:

$$\text{Max } 16x_1 + 15x_2 + 20x_3 - 18x_4$$

ST

$$2x_1 + x_2 + 3x_3 \leq 3000 \quad [1]$$

$$3x_1 + 4x_2 + 5x_3 - 60x_4 \leq 2400 \quad [2]$$

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$$x_2 \geq 200 \quad [4]$$

$$x_1 + x_2 + x_3 \geq 800 \quad [5]$$

$$x_1 - x_2 - x_3 = 0 \quad [6]$$

$$x_j \geq 0 \text{ for all } j$$

We assign a very large negative objective function coefficient, **-M**, (**+M** for minimization problem) to each artificial variable

We add artificial : **R4, R5, R6**, respectively to the fourth, fifth, and sixth equations.

Simplex method when some constraints are not “ $\leq$ ” constraints (cont.)

The solution

$$\text{Max } 16x_1 + 15x_2 + 20x_3 - 18x_4 - \text{MR4} - \text{MR5} - \text{MR6}$$

ST

$$2x_1 + x_2 + 3x_3 + s_1 = 3000 \quad [1]$$

$$3x_1 + 4x_2 + 5x_3 - 60x_4 + s_2 = 2400 \quad [2]$$

$$x_4 + s_3 = 32 \quad [3]$$

$$x_2 - s_4 + R_4 = 200 \quad [4]$$

$$x_1 + x_2 + x_3 - s_5 + R_5 = 800 \quad [5]$$

$$x_1 - x_2 - x_3 + R_6 = 0 \quad [6]$$

$$x_j \geq 0, s_j \geq 0, R_j \geq 0 \text{ for all } j$$

The simplex algorithm can then be used to solve this problem

Solving For the optimal solution of [**Maximization**] when there are artificial variables

Example # 1:

$$\text{MAX } 2x_1 + 5x_2$$

ST

$$x_1 \geq 4$$

$$x_1 + 4x_2 \leq 32$$

$$3x_1 + 2x_2 = 24$$

Solving For the optimal solution of **[Maximization]** when there are artificial variables (cont.)

The Solution

- By adding the appropriate slack, surplus, and artificial variables, we obtain the following:

$$\text{MAX } 2x_1 + 5x_2 - MR_1 - MR_3$$

ST

$$X_1 - s_1 + R_1 = 4$$

$$X_1 + 4x_2 + s_2 = 32$$

$$3x_1 + 2x_2 + R_3 = 24$$

$$X_1, x_2, s_1, s_2, R_1, R_3 \geq 0$$

Solving For the optimal solution of [Maximization] when there are artificial variables (cont.)

The initial table :

Basis	X1	X2	S1	S2	R1	R3	RHS
R1	1	0	-1	0	1	0	4
S2	1	4	0	1	0	0	32
R3	3	2	0	0	0	1	24
Z	-2	-5	0	0	+ M	+ M	0



R1, S2, R3 are basic variables.

- Make z consistent; (R1, R3) in z-row coefficient (+M,+M) it must be zero; By apply:

New z-row = old z-row + (-M * R1 row – M * R3 row)

} MAX objective function

New z-row = old z-row + (M * R1 row +M * R3 row)

} MIN objective function

Solving For the optimal solution of [Maximization] when there are artificial variables (cont.)

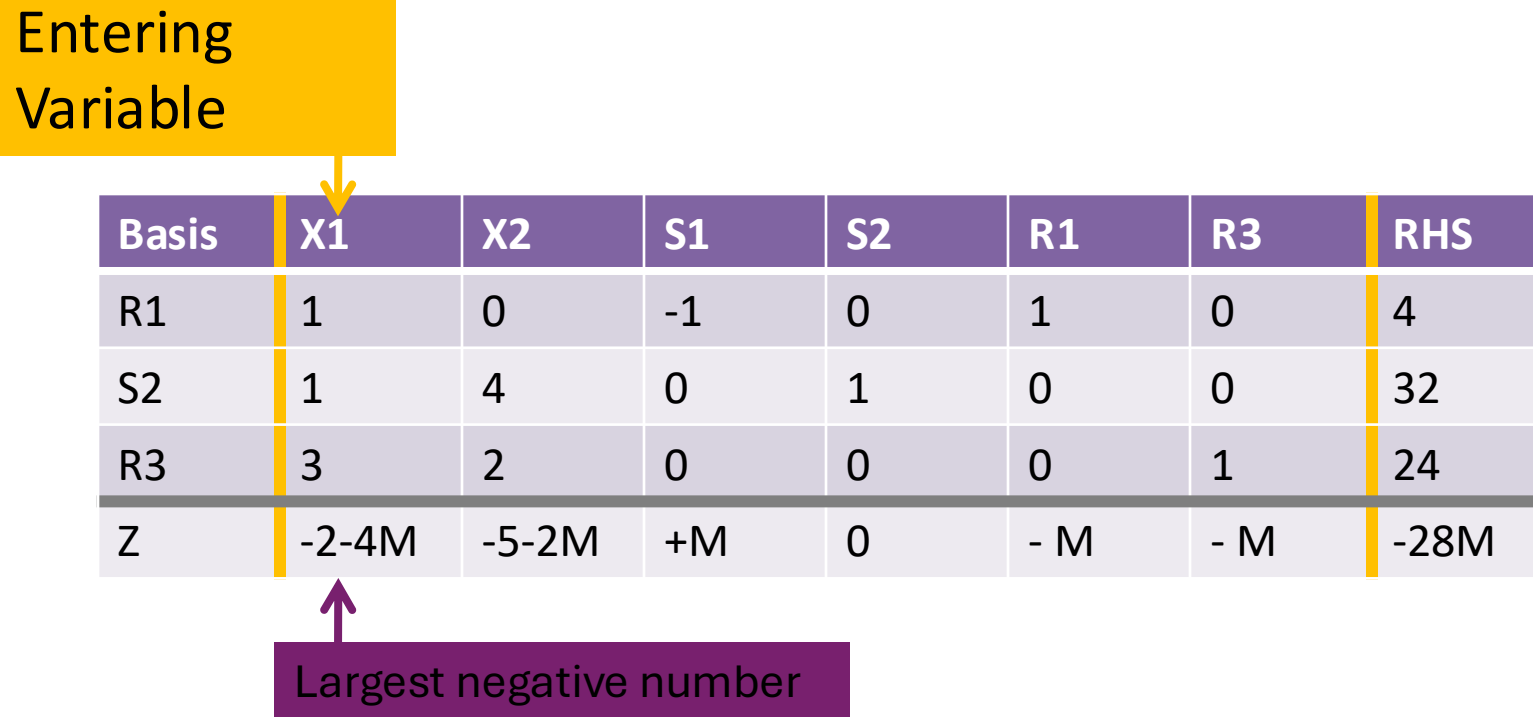
- Starting table:

Basis	X1	X2	S1	S2	R1	R3	RHS
R1	1	0	-1	0	1	0	4
S2	1	4	0	1	0	0	32
R3	3	2	0	0	0	1	24
Z	-2-4M	-5-2M	+M	0	- M	- M	-28M

Solving For the optimal solution of [Maximization] when there are artificial variables (cont.)

- To determine **Entering Variable**; We should look to the largest negative number in z-row.

Entering
Variable



Basis	X1	X2	S1	S2	R1	R3	RHS
R1	1	0	-1	0	1	0	4
S2	1	4	0	1	0	0	32
R3	3	2	0	0	0	1	24
Z	-2-4M	-5-2M	+M	0	- M	- M	-28M

Largest negative number

Solving For the optimal solution of [Maximization] when there are artificial variables (cont.)

- Calculate the ratio; then, determine the smallest positive number as **Leaving Variable**

Basis	X1	X2	S1	S2	R1	R3	RHS	Leaving Variable
R1	1	0	-1	0	1	0	4	4
S2	1	4	0	1	0	0	32	32
R3	3	2	0	0	0	1	24	8
Z	-2-4M	-5-2M	+M	0	-M	-M	-28M	

- Pivot element = $(1, 0, -1, 0, 1, 0, 4) / (1)$
 $(1, 0, -1, 0, 1, 0, 4)$

Solving For the optimal solution of [Maximization] when there are artificial variables (cont.)

- First iteration

Entering Variable								
Basis	X1	X2	S1	S2	R1	R3	RHS	Ratio
X1	1	0	-1	0	1	0	4	
S2	0	4	1	1	-1	0	28	28
R3	0	2	3	0	-3	1	12	4
Z	0	-5-2M	-2-3M	0	2+3M	-M	8-12M	

Leaving Variable

Solving For the optimal solution of [Maximization] when there are artificial variables (cont.)

- Second iteration

B	Entering Variable		S1	S2	R1	R3	RHS	Ratio
X1	1	2/3	0	0	0	1/3	8	12
S2	0	10/3	0	1	0	-1/3	24	7.2
S1	0	2/3	1	0	-1	1/3	4	6
Z	0	-11/3	0	0	0	2/3	+16	

Leaving Variable



Solving For the optimal solution of [Maximization] when there are artificial variables (cont.)

- Third iteration

Basis	X1	X2	S1	S2	R1	R3	RHS	Ratio
X1	1	0	-1	0	1	0	4	
S2	0	0	-5	1	5	-2	4	
X2	0	1	3/2	0	-3/2	1/2	6	
Z	0	0	11/3	0	-11/2	5/2	38	

Solving For the optimal solution of **[Maximization]** when there are artificial variables (cont.)

points	Classification	Reason
X1=0, X2=0	Not Feasible	R1, R3 both Positive (4, 24)
X1=4, X2=0	Not Feasible	R3 positive= 12
X1=8, X2=0	Feasible but not optimal	X2 is negative
X1=4, X2=6	Feasible and optimal	All $x_1, x_2 \geq 0$

Solving For the optimal solution of **[Minimization]** when there are artificial variables

Example # 2:

$$\text{Min } 4x_1 + x_2$$

ST

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Solving For the optimal solution of [**Minimization**] when there are artificial variables (cont.)

The Solution

- By adding the appropriate slack, surplus, and artificial variables, we obtain the following:

$$\text{Min } 4x_1 + x_2 + MR_1 + MR_2$$

ST

$$3x_1 + x_2 + R_1 = 3$$

$$4x_1 + 3x_2 - s_1 + R_2 = 6$$

$$x_1 + 2x_2 + s_2 = 4$$

$$x_1, x_2, s_1, s_2, R_1, R_2 \geq 0$$

Solving For the optimal solution of [Minimization] when there are artificial variables (cont.)

- The initial table:

Basis	X1	X2	S1	R1	R2	S2	RHS
R1	3	1	0	1	0	0	3
R2	4	3	-1	0	1	0	6
S2	1	2	0	0	0	1	4
Z	-4	-1	0	-M	-M	0	0

- New z-row = old z-row + (M * R1 row + M * R3 row)

Solving For the optimal solution of [Minimization] when there are artificial variables (cont.)

- Starting table:

Entering
Variable

Basis	X1	X2	S1	R1	R2	S2	RHS
R1	3	1	0	1	0	0	3
R2	4	3	-1	0	1	0	6
S2	1	2	0	0	0	1	4
Z	-4+7M	-1+4M	-M	0	0	0	9M

Leaving
Variable



Solving For the optimal solution of [Minimization] when there are artificial variables (cont.)

- First iteration

Entering Variable							Leaving Variable	
Basis	X1	X2	S1	R1	R2	S2	RHS	
X1	1	1/3	0	1/3	0	0	1	
R2	0	5/3	-1	-4/3	1	0	2	←
S2	0	5/3	0	-1/3	0	1	3	
Z	0	$(1+5M)/3$	-M	$(4-7M)/3$	0	0	$4+2M$	

Solving For the optimal solution of [Minimization] when there are artificial variables (cont.)

- Second iteration

Basis	X1	X2	S1	R1	R2	S2	
X1	1	0	1/5	3/5	-1/5	0	3/5
X2	0	1	-3/5	-4/5	3/5	0	6/5
S2	0	0	1	1	-1	1	1
Z	0	0	1/5	8/5 - M	-1/5 - M	0	18/5

Solving For the optimal solution of [Minimization] when there are artificial variables (cont.)

- Third iteration

Basis	X1	X2	S1	R1	R2	S2	RHS
X1	1	0	0	2/5	0	-1/5	2/5
X2	0	1	0	-1/5	0	3/5	9/5
s1	0	0	1	1	-1	1	1
Z	0	0	0	7/5 - M	-M	-1/5	17/5

- Optimal solution : $x_1 = 2/5$, $x_2 = 9/5$, $z = 17/5$

Simplex Algorithm – Special cases

- There are four special cases arise in the use of the simplex method.
 1. Degeneracy
 2. Alternative optima
 3. Unbounded solution
 4. Nonexisting (infeasible) solution

Simplex Algorithm – Special cases (cont.)

1. Degeneracy (no improve in objective)

- It typically occurs in a simplex iteration when in the minimum ratio test more than one basic variable determine 0, *hence two or more variables go to 0*, whereas only one of them will be leaving the basis.
- This is in itself not a problem, but making simplex iterations from a degenerate solution may give rise to cycling, meaning that after a certain number of iterations without improvement in objective value the method may turn back to the point where it started.

Simplex Algorithm – Special cases (cont.)

Example:

$$\text{Max } 3x_1 + 9x_2$$

ST

$$x_1 + 4x_2 \leq 8$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Simplex Algorithm – Special cases (cont.)

The solution:

- The constraints:

$$X_1 + 4x_2 + s_1 = 8$$

$$X_1 + 2x_2 + s_2 = 4$$

$$X_1, x_2, s_1, s_2 \geq 0$$

Simplex Algorithm – Special cases (cont.)

Entering Variable						Leaving Variable	
Basis	X1	X2	S1	S2	RHS		
s1	1	4	1	0	8	←	
s2	1	2	0	1	4		
Z	-3	-9	0	0	0		

Simplex Algorithm – Special cases (cont.)

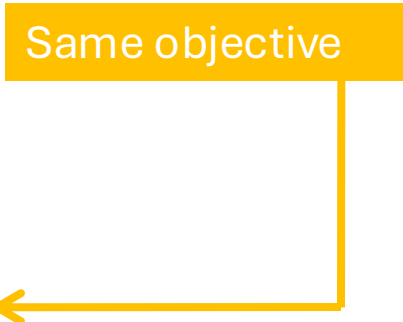
Entering
Variable

Basis	X1	X2	S1	S2	
X2	1/4	1	1/4	0	2
s2	1/2	0	-1/2	1	0
Z	-3/4	0	2/4	0	18

Leaving
Variable



Simplex Algorithm – Special cases (cont.)

Basis	X1	X2	S1	S2	RHS	Same objective 
X2	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	2	
X1	1	0	-1	2	0	
Z	0	0	$\frac{3}{2}$	$\frac{3}{2}$	18	

- Same objective no change and improve (cycle)
- It is possible to have no improve and no termination for computation.

Simplex Algorithm – Special cases (cont.)

2. Alternative optima

- If the z-row value for one or more nonbasic variables is 0 in the optimal tableau, alternate optimal solution exists.

Simplex Algorithm – Special cases (cont.)

Example:

$$\text{Max } 2x_1 + 4x_2$$

ST

$$x_1 + 2x_2 \leq 5$$

$$x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Simplex Algorithm – Special cases (cont.)

The solution

$$\text{Max } 2x_1 + 4x_2$$

ST

$$x_1 + 2x_2 + s_1 = 5$$

$$x_1 + x_2 + s_2 = 4$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Simplex Algorithm – Special cases (cont.)

Entering Variable				Leaving Variable	
Basis	X1	X2	S1	S2	
s1	1	2	1	0	4
s2	1	1	0	1	5
Z	-2	-4	0	0	0



Simplex Algorithm – Special cases (cont.)

- Optimal solution is 10 when $x_2=5/2$, $x_1=0$.

Basis	X1	X2	S1	S2	RHS
x2	1/2	1	1/2	0	5/2
s2	1/2	0	-1/2	1	3/2
Z	0	0	2	0	10

- How do we know from this tableau that alternative optima exist ?

Simplex Algorithm – Special cases (cont.)

- By looking at z-row coefficient of the nonbasic variable.

Entering Variable		X1	X2	S1	S2	Leaving Variable	
Basis							
x2		1/2	1	1/2	0	5/2	
s2		1/2	0	-1/2	1	3/2	
Z		0	0	2	0	10	



- The coefficient for x1 is 0, which indicates that x1 can enter the basic solution without changing the value of z.

Simplex Algorithm – Special cases (cont.)

- The second alternative optima is:

Basis	X1	X2	S1	S2	RHS
x2	0	1	1	-1	1
x1	1	0	-1	2	3
Z	0	0	2	0	10

- The new optimal solution is 10 when $x_1=3$, $x_2=1$

Simplex Algorithm – Special cases (cont.)

3. Unbounded solution

- It occurs when nonbasic variables are zero or negative in all constraints coefficient (max) and variable coefficient in objective is negative

Simplex Algorithm – Special cases (cont.)

Example

$$\text{Max } 2x_1 + x_2$$

ST

$$x_1 - x_2 \leq 10$$

$$2x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

Simplex Algorithm – Special cases (cont.)

The solution

Max $2x_1 + x_2$

ST

$$x_1 - x_2 + s_1 = 10$$

$$2x_1 + s_2 = 40$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Simplex Algorithm – Special cases (cont.)

Basis	X1	X2	S1	S2	RHS
x2	1	-1	1	0	10
x1	2	0	0	1	40
Z	-2	-1	0	0	0

- All value if x2(nonbasic variable) either zero or negative.
- So, solution space is unbounded

Simplex Algorithm – Special cases (cont.)

4. Infeasible solution

- R coefficient at end $\neq 0$
- This situation can never occur if all the constraints are of the type “ $\leq$ ” with nonnegative RHS